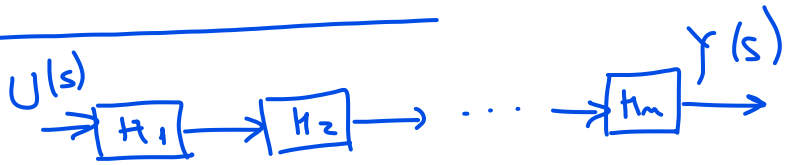
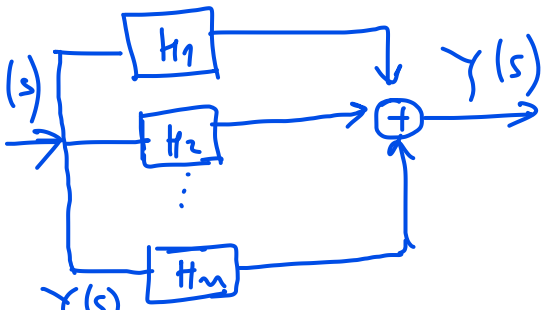
Conexiunea sistemelor

Legătura serie:



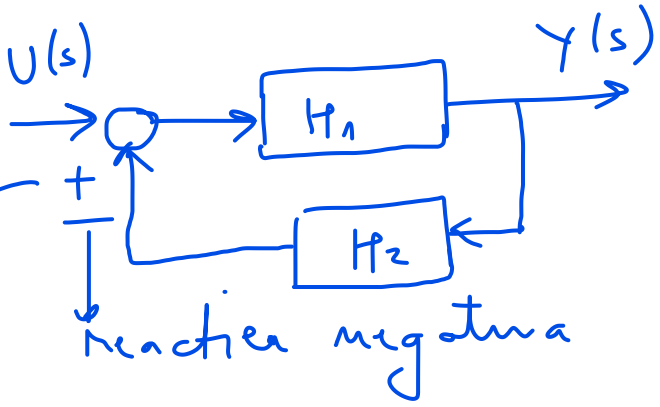
$$H_{serie}(s) = \prod_{i=1}^{m} H_i(s) = \frac{Y(s)}{U(s)}$$

Legătura paralel.



$$H_{paralel}(s) = \sum_{i=1}^{n} H_i(s) = \frac{Y(s)}{U(s)}$$

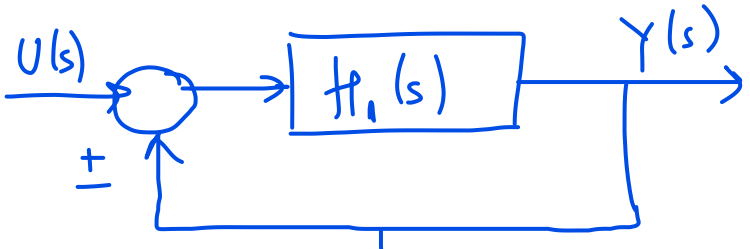
Legătura reactie



$$H_{reactie}(s) = \frac{H_1(s)\cdot H_2(s)}{1 \mp H_1(s)H_2(s)} = \frac{Y(s)}{U(s)}$$

+ semn contrar tipului de reactie

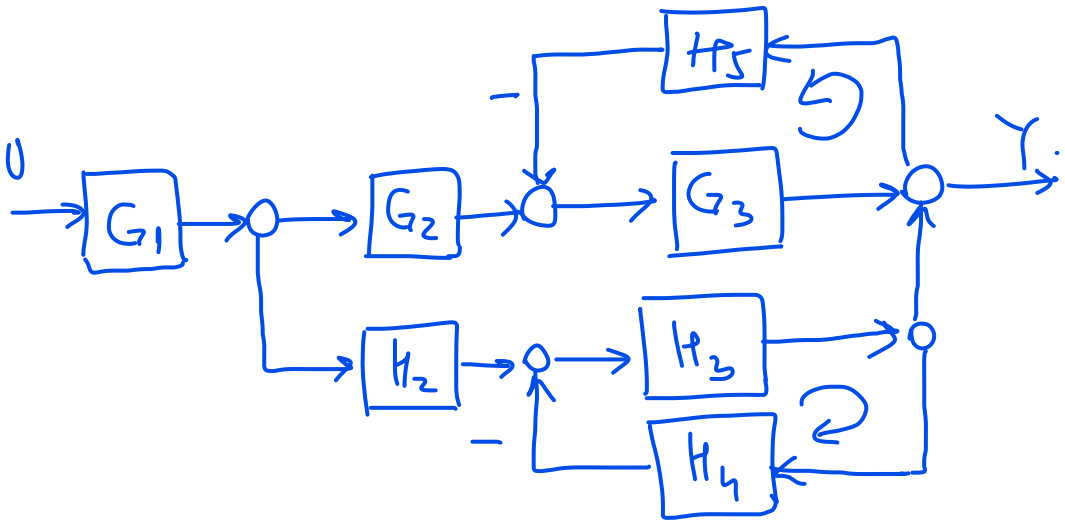
Obs



aici funcția de transfer ≡ Δ

$$\Rightarrow H_{reactie}(s) = \frac{H_1(s)}{1 \mp H_1(s)}$$

Regula de manipulare a funcțiilor de transfer a lui MASON.



Se cunosc:
G_1, G_2, G_3
H_2, H_3, H_4, H_5
Se cere $\frac{Y}{U}$

a) Căi directe de la U la Y.

$G_{uy_1} = G_1 \cdot G_2 \cdot G_3$

$G_{uy_2} = G_1 H_2 H_3$

b) Bucle $(j=1)$:

$P_1(1) = -G_3 H_5$

$P_2(1) = -H_3 H_4$

$P_3(1) = 0$

c) Perechi de bucle care nu se ating $(j=2)$

$P_1(2) = G_3 H_3 H_4 H_5$

$P_2(2) = 0$

d) Triplete de bucle care nu se ating $(j=3)$

$P_1(3) = 0$

$P_2(3) = 0$

e) Cvartete de bucle care nu se ating $(j=4)$

$P_1(4) = 0.$

Determinam Δ - parametru care caracterizează

graful schema

$$\Delta = 1 + (-1)^1 \cdot [P_1(1) + P_2(1)] + (-1)^2 \cdot P_1(2) + (-1)^3 \cdot P_1(3) + (-1)^4 \cdot P_1(4)$$

$$\Delta = 1 + G_3 H_5 + H_3 H_4 + G_3 H_3 H_4 H_5 + 0 + 0$$

$$\boxed{\Delta = 1 + G_3 H_5 + H_3 H_4 + G_3 H_3 H_4 H_5}$$

Determinam câte un coeficient de amplificare pentru fiecare cale directa de la U la Y.

$$\Delta_1 = 1 + (-1)^1 \cdot [P_2(1)] + (-1)^2 \cdot 0 + (-1)^3 \cdot 0 + (-1)^4 \cdot 0$$

↓

pt că nu atinge calea G_{uy_1}

$\Rightarrow \Delta_1 = 1 + H_3 H_4 \rightarrow$ coef. de amplificare cale directa G_{uy_1}

$$\Delta_2 = 1 + (-1)^1 \cdot [P_1(1)] + (-1)^2 \cdot 0 + (-1)^3 \cdot 0 + (-1)^4 \cdot 0$$

↓

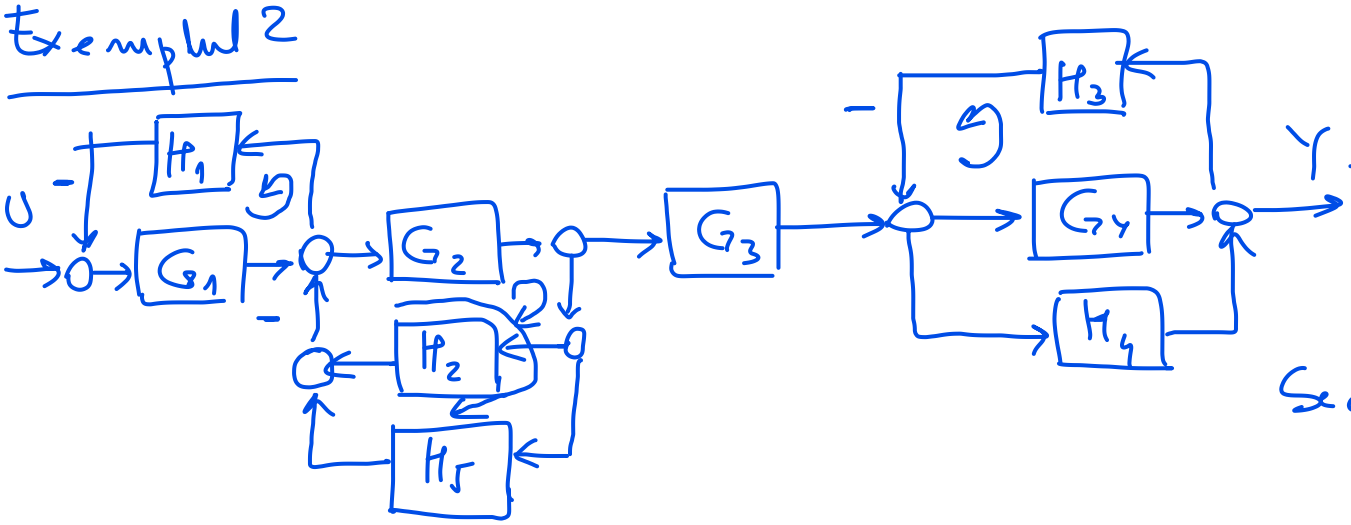
pt că nu atinge calea G_{uy_2}

$\Rightarrow \Delta_2 = 1 + G_3 H_5 \rightarrow$ coef. de amplificare cale directa G_{uy_2}

$$\frac{Y}{U} = \frac{G_{uy_1} \cdot \Delta_1 + G_{uy_2} \cdot \Delta_2}{\Delta}$$

$$\frac{Y}{U} = \frac{G_1 G_2 G_3 \cdot (1 + H_3 H_4) + G_1 H_2 H_3 (1 + G_3 H_5)}{1 + G_3 H_5 + H_3 H_4 + G_3 H_3 H_4 H_5}$$

Exemplul 2



Se cere $\frac{Y}{U}$

a) Căi directe de la U la Y

$G_{uy_1} = G_1 G_2 G_3 G_4$

$G_{uy_2} = G_1 G_2 G_3 H_4$

b) Bucle ($j=1$)

$P_1(1) = -G_1 H_1$ $\quad P_2(1) = -G_2 H_5$

$P_2(1) = -G_2 H_2$ $\quad P_4(1) = -G_4 H_3$

$P_5(1) = -H_3 H_4$

c) Perechi de bucle care nu se ating ($j=2$)

$P_1(2) = G_1 G_2 H_1 H_2$ $\quad P_3(2) = G_1 G_4 H_1 H_3$ $\quad P_5(2) = G_2 G_4 H_2 H_3$

$P_2(2) = G_1 H_1 G_2 H_2$ $\quad P_4(2) = G_1 H_1 H_3 H_4$ $\quad P_6(2) = G_2 H_2 H_3 H_4$

$P_7(2) = G_2 G_4 H_3 H_5$ $\quad P_8(2) = G_2 H_3 H_4 H_5$

d) Triplete de bucle care nu se ating ($j=3$)

$P_1(3) = -G_1 G_2 G_4 H_1 H_2 H_3$ $\quad P_2(3) = -G_1 G_2 H_1 H_2 H_3 H_4$

$$\Delta = 1 + (-1)^1 \cdot [P_1(1) + \dots + P_5(1)$$
$$+ (-1)^2 \cdot [P_1(2) \dots + P_3(2)]$$
$$+ (-1)^3 \cdot [P_1(3) + P_2(3)]$$

$$\Delta_1 = 1 + (-1)^1 \cdot [P_5(1)] + (-1)^2 \cdot 0 = 1 + H_3 H_4$$

$$\Delta_2 = 1 + (-1)^1 \cdot [P_4(1)] = 1 + G_4 H_3$$

$$\frac{Y}{U} = \frac{G_1 G_2 G_3 G_4 \Delta_1 + G_4 G_2 \Delta_2}{\Delta} =$$

$$= \frac{G_1 G_2 G_3 G_4 (1 + H_3 H_4) + G_1 G_2 G_3 H_4 (1 + G_4 H_3)}{\Delta}$$